

## CMSC 207: Chapter 9 Homework

May 10, 2026

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1. Prove that for all integers  $n$ ,  $k$ , and  $r$  for  $r \leq k \leq n$  that:

$$\binom{n}{k} \cdot \binom{k}{r} = \binom{n}{r} \cdot \binom{n-r}{k-r}.$$

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*Proof.*

**Lemma 1.** The factorial expansion of  $\binom{n}{r}$  is:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

By Lemma 1, we can rewrite the left-hand side of the equation as follows:

$$\begin{aligned} \binom{n}{k} \cdot \binom{k}{r} &= \left( \frac{n!}{k!(n-k)!} \right) \cdot \left( \frac{k!}{r!(k-r)!} \right) \\ &= \frac{n!k!}{k!r!(n-k)!(k-r)!} \\ &= \frac{n!}{r!(n-k)!(k-r)!}. \end{aligned}$$

By Lemma 1, we can rewrite the right-hand side of the equation as follows:

$$\begin{aligned} \binom{n}{r} \cdot \binom{n-r}{k-r} &= \left( \frac{n!}{r!(n-r)!} \right) \cdot \left( \frac{(n-r)!}{(k-r)!((n-r)-(k-r))!} \right) \\ &= \frac{n!(n-r)!}{r!(n-r)!(k-r)!((n-r)-(k-r))!} \\ &= \frac{n!}{r!(k-r)!(n-k)!}. \end{aligned}$$

Since multiplication is commutative under  $\mathbb{Z}$ , we have:

$$\frac{n!}{r!(n-k)!(k-r)!} = \frac{n!}{r!(k-r)!(n-k)!}.$$

Thus, the LHS and RHS of the equation are equal, so  $\binom{n}{k} \cdot \binom{k}{r} = \binom{n}{r} \cdot \binom{n-r}{k-r}$ . ■

2. The binomial theorem states that for any numbers  $a$  and  $b$ :

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k \text{ for any integer } n \geq 0.$$

Use this theorem to show that for any integer  $n \geq 0$ :

$$\sum_{k=0}^n (-1)^k \binom{n}{k} 3^{n-k} 2^k = 1.$$

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*Proof.*

By the binomial theorem, we have:

$$(3 - 2)^n = \sum_{k=0}^n \binom{n}{k} 3^{n-k} (-2)^k.$$

Since  $3 - 2 = 1$ , we have:

$$1^n = \sum_{k=0}^n \binom{n}{k} 3^{n-k} (-2)^k.$$

Since  $1^n = 1$  for all integers  $n \geq 0$ , we have:

$$\sum_{k=0}^n \binom{n}{k} 3^{n-k} (-2)^k = 1.$$

Since  $(-2)^k = (-1)^k \cdot 2^k$  for all integers  $k \geq 0$ , we have:

$$\sum_{k=0}^n \binom{n}{k} 3^{n-k} (-1)^k \cdot 2^k = \sum_{k=0}^n (-1)^k \binom{n}{k} 3^{n-k} \cdot 2^k.$$

Thus, we have shown that:

$$\sum_{k=0}^n (-1)^k \binom{n}{k} 3^{n-k} 2^k = 1. \quad \blacksquare$$